Understanding Enums

```
enum Direction {
    NORTH,
    SOUTH,
    EAST,
    WEST;
}
```

```
class Direction extends Enum<Direction>{
}
```

- java.lang.Enum overrides ✓
- Each Constants are static & Final & are actually objects of class Direction.

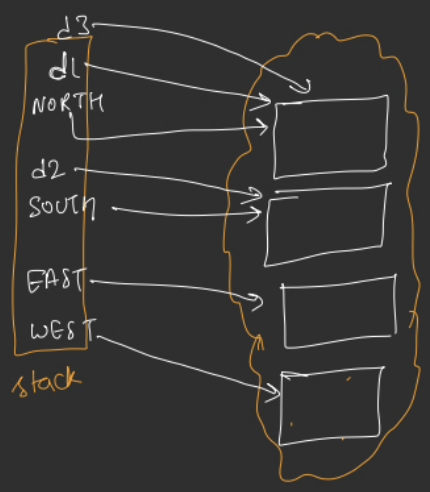
```
enum Direction {
    NORTH,
    SOUTH,
    EAST,
    WEST;
}
```

- → Each Enum is class
- → Extends Enum class.
- → Each Constants are static & Final Objects of type Direction.

```
final class Direction extends Enum<Direction>{
    Public static final Direction NORTH = new Direction();
    Public static final Direction SOUTH = new Direction();
    Public static final Direction EAST  = new Direction();
    Public static final Direction WEST  = new Direction();

    private Direction() {
    }
}
```

Marker

```
main()
Direction d1 = Direction.NORTH;
Direction d2 = Direction.SOUTH;
d3 = ——.NORTH.
```

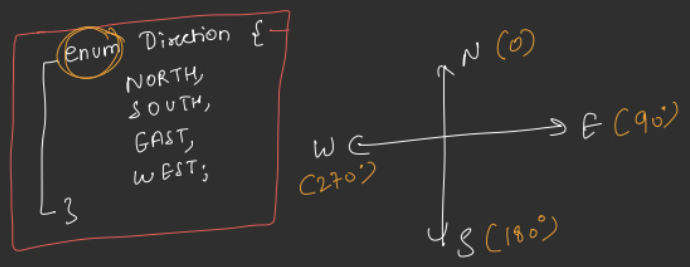


Day of the week

- Monday
- Tuesday
- ⋮
- Sunday

```
enum DayOfWeek {
    MONDAY,
    TUESDAY,
    ⋮
    SUNDAY;
}
```

Enum → Class
- States (variables)
- Behaviours

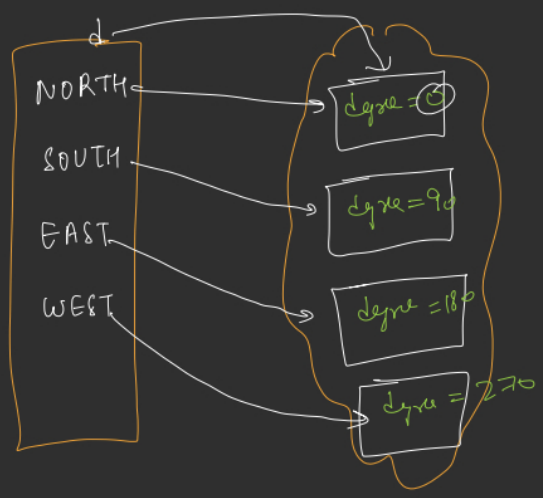



```java
enum Direction {

    NORTH(0),
    EAST(90),
    SOUTH(180),
    WEST(270);

    private int degrees;

    Direction(int degrees) {
        this.degrees = degrees;
    }

    public int getDegrees() {
        return degrees;
    }
}
```

(our code)

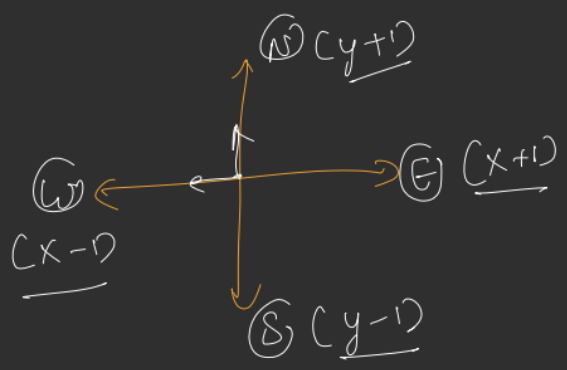
```java
final class Direction extends Enum<Direction> {

    public static final Direction NORTH = new Direction(0);
    public static final Direction EAST = new Direction(90);
    public static final Direction SOUTH = new Direction(180);
    public static final Direction WEST = new Direction(270);

    private int degrees;

    private Direction(int degrees) {
        this.degrees = degrees;
    }

    public int getDegrees() {
        return degrees;
    }
}
```

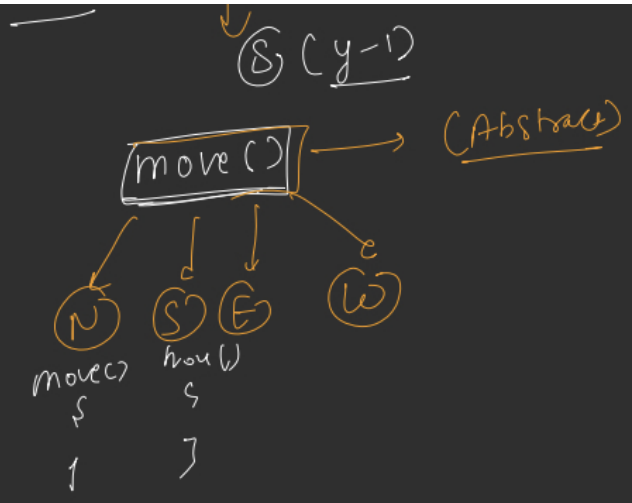
compiler generated



Direction d = Direction.NORTH;

d.getDegree() → 0



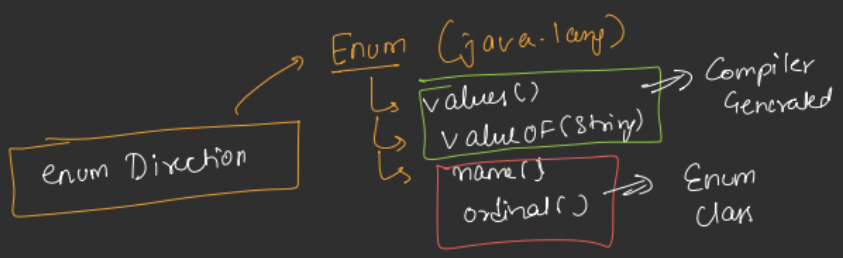



```java
enum Direction {

    NORTH {
        @Override
        public void move() {
            System.out.println("Moving Up");
        }
    },

    SOUTH {
        @Override
        public void move() {
            System.out.println("Moving Down");
        }
    },

    EAST {
        @Override
        public void move() {
            System.out.println("Moving Right");
        }
    },

    WEST {
        @Override
        public void move() {
            System.out.println("Moving Left");
        }
    };

    public abstract void move();
}
```

```java
final class Direction extends Enum<Direction> {

    public static final Direction NORTH = new Direction() {
        @Override
        public void move() {

        }
    };
    public static final Direction EAST = new Direction() {
        @Override
        public void move() {

        }
    };
    public static final Direction SOUTH = new Direction(){
        @Override
        public void move() {

        }
    };
    public static final Direction WEST = new Direction(){
        @Override
        public void move() {

        }
    };


    public abstract void move();
}
```

ENUM

↳ Existing functionality !



ENUM

↳ Existing functionality !

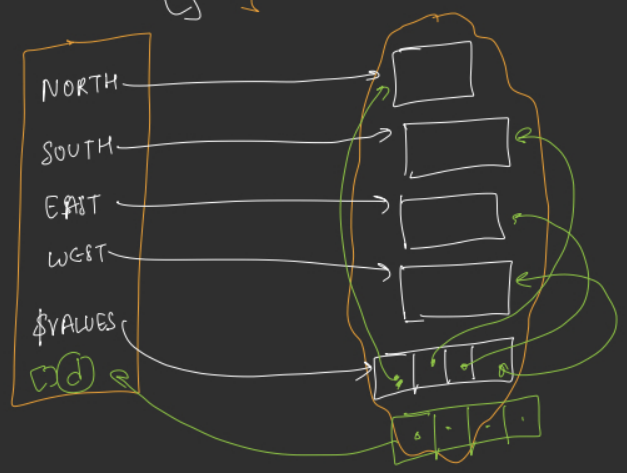
enum Direction → Enum (java.lang)

- ↳ values() → Compiler Generated
- ↳ valueOF(String)
- ↳ name()
- ↳ ordinal() → Enum Class

```
enum Direction {
        NORTH,
        SOUTH,
        EAST,
        WEST;
}
```

⇒

```
final class Direction extends Enum<Direction>{
    ____________
    ____________
    ____________
    ____________

    private static final Direction[] $VALUES = {
        NORTH,
        SOUTH,
        EAST,
        WEST;
    }

    public static Direction[] values()
        return $VALUES.clone();

    public static Direction valueOf(String s)
        super.valueOf(Direction.class, s);
}
```



valueOF()

Direction d = Direction.valueOf("EAST");

Enums

- ↳ Day of the week
- ↳ payment status.
- ↳ Order status.
- ↳ logs (INFO, ERROR, DEBUG)